

David Colindres

www.linkedin.com/in/DR1PD | www.github.com/DR1PD | david.colindres@live.com



Education

The University of Oklahoma; Norman, OK

GPA: 4.00/4.00

M.S./B.S., Industrial & Systems Engineering and M.A., Econometrics

Expected Dec 2026

Minors; Engineering Leadership, Mathematics, International Studies, and Global Energy, Environment & Resources

Thesis: Classification in Machine Learning for Financial Fraud Detection (Quant Risk, Neural Networks, Deep Learning, & Class Imbalance)

Relevant Coursework: Stochastic Calculus & Markov Chains, Linear Algebra, Ordinary Differential Equations, Optimization, Probabilistic Models, Monte Carlo Simulation, Micro/Advanced/Financial Econometrics, Machine Learning

Skills

Programming; Python (Pandas, NumPy, CVXPY, scikit-learn, PyTorch, statsmodels), R, SQL, C, JavaScript, Git, $\LaTeX$, Jupyter

Quantitative; Portfolio Optimization, Covariance Estimation (Random Matrix Theory, Ledoit-Wolf), Hidden Markov Models,

Walk-Forward Backtesting, VaR/CVaR, Time-Series Forecasting, Regression (OLS, IV), DID, Machine Learning (Boosting, Loss Fns)

Tools; MATLAB, Excel (Advanced), RStudio, Stata, Tableau, AWS, Azure

Quantitative Research

Multi-Period Portfolio Optimization with Transaction Costs | Python, CVXPY

- Solved [X] monthly rebalances across [X] years of daily equity data vs. equal-weight & S&P 500 benchmarks by building multi-period portfolio optimizer in CVXPY with transaction-cost modeling
- Identified forecast quality over transaction costs as the main limit on active trading performance by conducting cost/risk-aversion sensitivity analysis across [X] backtests

Eigenportfolios & Random Matrix Theory Covariance Cleaning | Python, NumPy

- Enhanced covariance estimation across [X] equities via Random Matrix Theory (Marchenko-Pastur) denoising, benchmarked vs Ledoit-Wolf & sample covariance in min-variance backtests
- Verified consistent findings for data-driven covariance estimation by stress-testing results across [X] window/universe configurations & sub-universes

Hidden Markov Model Regime Detection | Python, hmmlearn, hand-rolled forward algorithm

- Detected the **2008 & 2020** crises within [X] trading days by building Gaussian HMM regime-switching strategy with walk-forward backtesting on [X] years of daily data
- Fixed lookahead/survivorship biases & log-return bug that flipped headline results by auditing full codebase ([X] findings)—documenting limitations for transparent/reproducible research

Relevant Experience

Cloud/Software Engineering Intern | Federal Reserve Bank of Chicago | Chicago, IL

Jun 2026 – Aug 2026

- Optimized cloud spend allocation across **1000+** AWS accounts via constrained portfolio optimization (CVXPY) under stochastic usage
- Cut cost oversight from month-end to same-day by automating budget alerts in Python, auto-routing to FinOps product owners

Cloud Architecture Intern | Accenture plc | Oklahoma City, OK

Jun 2025 – Aug 2025

- Advanced **\$2B+** migration by delivering multi-cloud architecture across AWS/Azure, aligning IaaS/PaaS models & CI/CD pipelines
- Accelerated enterprise adoption workflows by automating cloud-native tooling with JavaScript to streamline API reviews

Junior Investment Analyst | Crimson Prairie Ventures | Norman, OK

Jan 2024 – May 2025

- Placed 1st in Devon Energy VC Case Competition, building valuation framework modeling cost, cashflow & **~75%** time savings
- Guided **~\$50,000** in annual capital allocation via financial models for pre-seed startup due diligence, evaluating risk & return

Process Engineering Intern; former Associate | Walmart Automation | Douglas, GA

Jun 2022 – Aug 2024

- Saved **~\$67,000** in costs & reduced error to **~0.38%** by forecasting econometric models for applied quality assurance

Service & Leadership

Founder | OU Quant — Student Quantitative Research Pod

Jun 2026 – Present

- Established OU's first student quant research group & lead flagship research on contango patterns in futures/commodities

Graduate Chair; Holmes Leadership Associate | Jerry Holmes Leadership Program for Engineers & Scientists

Apr 2022 – Present

- Broke three **10+** year program records—new members **168%** above average, **2x**'d alumni mentor program & LinkedIn outreach

Graduate Residence Coordinator; Honors Resident Advisor | Graduate Assistantship

Jan 2023 – Present

- Cultivate leadership/community for **24** staff & **~2000+** residents via frequent programming, mentoring & allocating resources

TEDx Speaker; Public Service Administrator | OU Love Letters

May 2022 – Mar 2026

- Delivered TEDxOU talk on anonymous expression; grew to **4,500+** followers curating **6,000+** submissions—featured in OU Daily (x2)

Study Abroad Ambassador | Arezzo, IT/Clermont, FR/Puebla, MX

Fall '22, Summer '22/'24/'25

- Participated in Study Abroad across **4** academic terms in France/Italy/Mexico for engineering, supply chain & sustainability coursework

Honors & Certificates

K. Partridge Campus Life Award, 2025

Hackathon Winner (1x), 2026

Lean/Six Sigma Green Belt (ICGB), 2023

Top 4 IISE National Capstone Finalist, 2025

President's Leadership Class (Top 1%), 2021-2025

RA Student Staff Member of the Year, 2023-2024